

Homework 01: Moneyball Data Analysis & Predictive Modeling Report

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1. Introduction

The objective of this study is to construct a multiple linear regression model capable of predicting the total number of wins (`TARGET_WINS`) achieved by Professional Baseball teams during a regular season. The dataset spans historical team performance records from 1871 through 2006. Each observation represents a single season for a team and contains aggregated metrics covering offensive hitting, baserunning, pitching, and defensive fielding performance.

The analysis utilizes two distinct datasets:

- **Training Dataset:** Contains 2,276 team-season observations with historical statistics and the target variable `TARGET_WINS`.
- **Evaluation Dataset:** Contains 259 team-season observations with identical predictor fields, where the actual win counts are withheld for out-of-sample model evaluation.

By leveraging exploratory data analysis, data cleaning, feature transformation, and iterative model validation, this report presents a parsimonious linear regression model that accounts for domain-specific baseball dynamics while controlling for severe multicollinearity.

2. Exploratory Data Analysis

2.1 Target Variable Characteristics

The dependent variable, `TARGET_WINS`, reflects a 162-game season structure across modern observations.

- The target variable exhibits an approximately normal distribution centered around a sample mean of 80.79 wins (≈ 80.8) and a median of 82.00 wins.

- The distribution spans from a minimum of 0 wins to a maximum of 146 wins.
- Extreme low values (<20 wins) represent historical short seasons from the 19th century or data collection artifacts, warranting careful handling during pre-processing.

2.2 Missing Data Audit

An evaluation of missingness across all 16 candidate features in the training dataset revealed significant missing values in several metrics:

- **TEAM_BATTING_HBP (Hit by Pitch):** Missing 2,085 observations (91.61%). Excluded entirely from modeling due to excessive missingness.
- **TEAM_BASERUN_CS (Caught Stealing):** Missing 772 Observations (33.92%). Addressed by creating a missingness indicator flag prior to median imputation.
- **TEAM_FIELDING_DP (Double Plays):** Missing 286 observations (12.57%). Addressed by creating a missingness indicator flag prior to median imputation.
- **TEAM_BASERUN_SB (Stolen Bases):** Missing 131 observations (5.76%). Imputed using training set medians.
- **TEAM_BATTING_SO (Strikeouts):** Missing 102 observations (4.48%). Imputed using training set medians.
- **TEAM_PITCHING_SO (Pitching Strikeouts):** Missing 102 observations (4.48%). Imputed using training set medians.

2.3 Feature Associations & Multicollinearity

Bivariate correlation analysis revealed strong logical relationships, alongside key modeling challenges:

- Positive associations exist between offensive hitting statistics and total wins.
- Extreme collinearity is present between corresponding offensive and pitching metrics. Specifically, batting home runs and pitching home runs allowed near perfect linear dependence, introducing variance inflation when included together in standard Ordinary Least Squares estimation.

3. Data Preparation & Feature Engineering

To prepare the dataset for statistical modeling, a systematic data preparation pipeline was implemented:

1. **Exclusion of High-Missingness Features:** TEAM_BATTING_HBP was permanently dropped due to missing over 91% of its observations.
2. **Missing Value Imputation:** For predictors with < 10% missingness, missing values were imputed using training set medians. For predictors missing > 10% observations, missingness binary indicator flags were created prior to median imputation to retain structural missingness information.
3. **Derivation of Single Base Hits:** Total hits inherently includes extra-base hits. To isolate independent hitting effects and eliminate mathematical redundancy, single base hits were calculated as:

$$\text{TEAM_BATTING_1B} = \text{TEAM_BATTING_H} - (\text{TEAM_BATTING_2B} + \text{TEAM_BATTING_3B} + \text{TEAM_BATTING_HR})$$

4. **Outlier Mitigation (Winsorization):** To limit the distorting impact of historical statistical anomalies, continuous predictors were capped at the 1st and 99th percentile bounds.
5. **Logarithmic Transformations:** Right-skewed defensive variables showing non-linear relationships most notably fielding errors were log-transformed into LOG_TEAM_FIELDING_E to promote homoscedasticity and residual normality.

4. Model Building & Evaluation

Three candidate OLS linear regression models were specified and evaluated using cross-validation and goodness-of-fit metrics:

Model Performance Overview

- **Model 1 (Full Baseline):** Utilizes all 16 candidate predictors. It yields a Multiple R^2 of 0.3220, an Adjusted R^2 of 0.3170, and a Root Mean Squared Error (RMSE) of 12.980 wins. However, it suffers from severe multicollinearity, with maximum Variance Inflation Factor (VIF) values exceeding 50.0 on home run and strikeout variables.
- **Model 2 (Reduced Feature Model):** Reduces complexity to 14 predictors by pruning non-significant terms like pitching home runs allowed (TEAM_PITCHING_HR) and caught stealing (TEAM_BASERUN_CS). It achieves a Multiple R^2 of 0.3210, an Adjusted R^2 of 0.3170, and an RMSE of 13.010 wins. Most retained features are statistically significant ($p < 0.05$), though maximum VIF remains elevated above 45.0.

- **Model 3 (Derived Differential Model):** Streamlines the specification to 11 predictors by replacing correlated raw pairs with net performance differentials. It achieves a Multiple R^2 of 0.3120, an Adjusted R^2 of 0.3096, and an RMSE of 13.089 wins. Every retained predictor is statistically significant ($p < 0.05$), and maximum VIF drops to 9.90, successfully meeting the multicollinearity threshold of < 10.0 .

Detailed Specifications

- **Model 1 Details:** Serves as the benchmark using all pre-processed and engineered predictors. Despite solid explanatory capacity, near-perfect correlations between offensive and defensive metrics inflate standard errors.
- **Model 2 Details:** Eliminates redundant noise variables while retaining explanatory power comparable to Model 1. However, structural collinearity between raw hitting and pitching totals persists.
- **Model 3 Details:** Replaces collinear offensive and defensive pairs with combined differential and transformed features:
 - **Net Walks:** $NET_BB = TEAM_BATTING_BB - TEAM_PITCHING_BB$
 - **Net Strikeouts:** $NET_SO = TEAM_BATTING_SO - TEAM_PITCHING_SO$
 - **Log Pitching Hits Allowed:** $LOG_TEAM_PITCHING_H$

5. Conclusion & Model Selection

Final Model Selection Rationale

Model 3 is selected as the optimal deployment model. Although Model 1 displays a slightly higher unadjusted R^2 (0.322 vs. 0.312), Model 3 successfully solves severe multicollinearity:

1. **Multicollinearity Resolution:** All predictor VIF values in Model 3 fall strictly below the threshold of 10.0 (maximum VIF = 9.90), ensuring stable coefficient estimates and reliable hypothesis testing.

2. **Statistical Significance:** All retained features are statistically significant at the $\alpha=0.05$ level.
3. **Generalization:** Maintains robust predictive capacity with a Root Mean Squared Error (RMSE) of 13.089 wins.

Residual Diagnostics & Limitations

Diagnostic checks on Model 3 residuals confirm that error terms are approximately zero-centered with consistent variance across predicted values. However, inspection of the Normal Q-Q plot shows heavy tail deviations at the extreme ends. These tail anomalies correspond to early historical seasons (1870s–1890s) where shorter schedules and non-standardized scoring distorted team record aggregates.

Out-of-Sample Evaluation Predictions

Applying Model 3 to the 259 unlabelled records in the Evaluation dataset yields the following prediction profile:

- **Mean Predicted Wins:** 80.34 wins
[PDF](#)
- **Median Predicted Wins:** 80.91 wins
[PDF](#)
- **Minimum Predicted Wins:** 41.76 wins
[PDF](#)
- **Maximum Predicted Wins:** 109.52 wins
[PDF](#)

The distribution of out-of-sample predictions aligns with expected regular-season win totals, confirming model stability when generalizing to unseen evaluation data.

6. Appendix

- **Source Code:**

[http://localhost:8888/notebooks/Downloads/hw01_moneyball_analysis_python_final%20report.i
pynb?](http://localhost:8888/notebooks/Downloads/hw01_moneyball_analysis_python_final%20report.ipynb?)